

# 企業資訊通訊與網路 期末考古題：題目與解答

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說明：本文件收錄三份期末考卷（題目＋參考解答）。解答依課本投影片 ch06–ch11 內容整理（藍字）。

已刪除「文字完全相同」的重複題：(1) ADSL 填充題（試卷二原 A7，與試卷一 A10 相同）；(2) 無線傳播問題簡答（試卷一原 B2，與試卷二 B2 相同，保留較完整的試卷二版本）。同考點但數字／措辭不同者均保留。

## 試卷一：企業資訊通訊與網路 期末考 (109-1)

### A. 填充 (Fill in the Blanks) (30%)

1. "fe80::c2f:a4:300" is the simplified representation of IPv6 address: \_\_\_\_.

【解答】 fe80:0000:0000:0000:0c2f:00a4:0300 (:: 代表 4 組全零；展開為完整 8 組)。  
[ch08 p77–82]

2. 10.34.65.117 bitwise-AND 255.255.240.0 = \_\_\_\_.

【解答】 10.34.64.0 (65 & 240 = 64, 第 4 octet & 0 = 0)。 [ch08 p19–21,55–58]

3. Signal strength at 2 m is 80 mW; at 8 m it would be \_\_\_\_ mW.

【解答】 5 mW。  $S_2 = 80 \times (2/8)^2 = 80 \times 1/16 = 5$ 。 [ch06 p14–16]

4. \_\_\_\_ is transmission with channel bandwidth much wider than required for the signal's speed.

【解答】 Spread spectrum transmission (展頻傳輸；2.4/5 GHz 頻段法規要求，可降低多路徑干擾)。 [ch06 p45]

5. When an IPv6 packet is forwarded by a router, the \_\_\_\_ field is decreased by one.

【解答】 Hop Limit (功能等同 IPv4 的 TTL)。 [ch08 p88]

6. \_\_\_\_ is a protocol, usually used by routers, to send error or supervisory messages.

【解答】 ICMP (Internet Control Message Protocol)。 [ch09 p69–71]

7. In the \_\_\_\_ mode of IPsec, only the two IPsec gateways need digital certificates.

【解答】 Tunnel (隧道模式；只有兩端 gateway 需憑證與設定)。 [ch09 p50–52]

8. To connect all sites via PSDN, each site uses a \_\_\_\_ line as access line to ISP's nearest POP.

【解答】 Leased line (專線；當作 access line 連到最近的 POP)。 [ch10 p37]

9. \_\_\_\_ is a routing mechanism. It inserts a label in front of the IP header.

【解答】MPLS (Multiprotocol Label Switching ; 以 label 查表轉送, 較傳統路由便宜)。〔ch10 p52–56〕

10. \_\_\_\_ provides residential Internet access over one-pair voice-grade UTPs, high downstream, limited upstream.

【解答】ADSL (Asymmetric DSL ; 利用既有 1-pair VG UTP)。〔ch10 p12〕

## B. 簡答 (Short Answer)

1. 802.11 operates in 2.4GHz and 5 GHz bands.

(1). Main advantage of 2.4GHz operation?

(2). Main advantage of 5GHz operation?

【解答】(1) 傳播特性較佳 (涵蓋距離較遠、穿透障礙物較好)。 (2) 頻寬較多, 可有 11–24 個非重疊通道, 附近 AP 易使用不重疊通道 (干擾少)、速度較高。〔ch06 p34,37,38〕

2. Your company uses private IP addresses 10.x.x.x.

(1). For at least 50 subnets, how many bits for the subnet part?

(2). How many hosts per subnet?

【解答】(1) 6 bits ( $2^6 - 2 = 62 \geq 50$  ; 5 bits 只有 30 不夠)。 (2) 10.x.x.x 為 /8, 剩 24 bits, 扣 6 bits subnet  $\rightarrow$  host = 18 bits  $\rightarrow 2^{18} - 2 = 262,142$  hosts。〔ch09 p5–8〕

3. BitTorrent P2P file retrieval.

(1). What is a tracker?

(2). What is a swarm?

【解答】(1) Tracker : 協助節點找到其他節點的伺服器, 記錄哪些 peer 在分享該檔、各擁有哪些片段。 (2) Swarm : 正在共同分享／下載同一檔案的所有 peer (種子+下載者) 的集合。〔ch11 P2P〕

4. CSMA/CA+ACK is the MAC scheme used by 802.11 WLANs.

(1). What does “Carrier Sense Multiple Access” mean?

(2). CA means Collision Avoidance. How can collisions be avoided?

【解答】(1) 傳送前先「聆聽載波」, 若偵測到別人正在傳就等待, 藉此控制多台裝置不同時傳送。 (2) 當前發送者停止後, 等待中的各裝置先各自等一段「隨機」時間再傳, 避免同時搶送造成碰撞。〔ch06 p59–60〕

5. Explain the following header fields.

(1). Sequence Number (in TCP)

(2). TTL (in IPv4)

(3). Next Header (in IPv6)

【解答】(1) Sequence Number : 每個 TCP segment 的序號, 接收端用來重組成原訊息、偵測遺失。 (2) TTL : 發送端設值(64–128), 每經一台 router 減 1, 減到 0 即丟棄(防止封包無限

繞行)。(3) Next Header：指出下一個標頭的型態—延伸標頭或上層協定 (TCP=6,UDP=17,ICMPv6=58), 延伸標頭以此欄串接。〔ch08 p70,93–97,100〕

6. PSK and 802.1X are the two 802.11i modes.

(1). Which is better for a company with many access points?

(2). How does the PSK mode work?

【解答】(1) 802.1X 模式（使用集中式認證伺服器，適合多 AP 企業）。(2) PSK：在 AP 與所有用戶端設定同一把預共享金鑰(由  $\geq 20$  字元 passphrase 產生)，知道 PSK 即通過認證；適合家用/單一 AP 的小環境。〔ch07 p13–16〕

7. The following is a routing table. Suppose the metric is speed.

| Row | Destination  | Subnet mask   | Metric (speed) | Interface | Next Hop      |
|-----|--------------|---------------|----------------|-----------|---------------|
| 1   | 0.0.0.0      | 0.0.0.0       | 6              | 1         | 163.22.1.254  |
| 2   | 156.0.0.0    | 255.0.0.0     | 10             | 6         | 156.1.1.254   |
| 3   | 156.63.0.0   | 255.255.0.0   | 12             | 5         | 156.63.1.254  |
| 4   | 156.63.0.0   | 255.255.0.0   | 18             | 8         | 156.19.1.254  |
| 5   | 156.110.0.0  | 255.255.0.0   | 12             | 9         | 156.22.2.1    |
| 6   | 156.110.45.0 | 255.255.255.0 | 25             | 2         | local         |
| 7   | 156.114.0.0  | 255.255.0.0   | 12             | 3         | 156.25.1.254  |
| 8   | 156.114.24.0 | 255.255.255.0 | 21             | 7         | local         |
| 9   | 156.114.26.0 | 255.255.255.0 | 36             | 4         | 156.114.2.1   |
| 10  | 156.114.26.0 | 255.255.255.0 | 30             | 3         | 156.114.3.252 |

(1). dest = 156.114.26.9 ? (2). dest = 156.110.45.7 ? (3). dest = 156.63.1.2 ?

【解答】(1) 符合 Row 1,2,7,9,10；最長前綴 9、10 皆 /24，比速度  $36 > 30 \rightarrow$  選 Row 9。

【解答】(2) 符合 Row 1,2,5,6；最長前綴 Row 6 為 /24  $\rightarrow$  選 Row 6。

【解答】(3) 符合 Row 1,2,3,4；最長前綴 3、4 皆 /16，比速度  $18 > 12 \rightarrow$  選 Row 4。〔ch08 p34–48〕

8. (1) SMTP commands (a)(b)(c) for John $\rightarrow$ Mary; (2) HTTP request (d)(e) for http://abc.com/news.html.

【解答】(1) (a)=MAIL（組成 MAIL FROM:）、(b)=RCPT（組成 RCPT TO:）、(c)=.（單獨一個句點，即 <CRLF>.<CRLF> 結束訊息）。(2) (d)=GET（GET /news.html HTTP/1.1）、(e)=Host（Host: abc.com）。〔ch11 p19–26,51〕

9. SIP and RTP are signaling and transport protocols of VoIP.

(1). How does a caller make a call to a callee via two proxy servers?

(2). VoIP uses RTP over UDP. Why UDP instead of TCP?

【解答】(1) 撥號端送 SIP INVITE 給自己的(第一台)proxy，轉送到對方的(第二台)proxy，再轉給被叫端；回應沿反向路徑回來。信令建立連線後，RTP 語音直接在主叫與被叫端之間傳送。(2) 語音是即時的，沒有時間等待重送遺失封包；UDP 輕量、開銷小，適合即時流量；RTP 再於 UDP 上加序號與時間戳。〔ch11 p41–44〕



## 試卷二：企業資訊通訊與網路 期末考 (2020)

### A. 填充 (Fill in the Blanks) (30%)

1. "2001::1:fe20:a42:3e" is the simplified representation of IPv6 address: \_\_\_\_.

【解答】2001:0000:0000:0000:0001:fe20:0a42:003e (:: 代表 3 組全零)。 [ch08 p77–82]

2. 163.22.72.51 bitwise-AND 255.255.224.0 = \_\_\_\_.

【解答】163.22.64.0 (72 & 224 = 64, 第 4 octet & 0 = 0)。 [ch08 p19–21,55–58]

3. Signal strength at 3 m is 80 mW; at 12 m it would be \_\_\_\_ mW.

【解答】5 mW。  $80 \times (3/12)^2 = 80 \times 1/16 = 5$ 。 [ch06 p14–16]

4. The subnet part is 6 bits long. Company A may have at most \_\_\_\_ subnets.

【解答】62 個 ( $2^6 - 2 = 62$  ; 扣掉全 0 與全 1)。 [ch09 p6]

5. When an IPv4 packet is forwarded by a router, the \_\_\_\_ field is decreased by one.

【解答】TTL (Time to Live)。 [ch08 p70]

6. CSMA/CA+ACK is mandatory; another optional MAC mechanism \_\_\_\_ can solve the hidden node problem.

【解答】RTS/CTS (Request-to-Send / Clear-to-Send)。 [ch06 p69]

7. \_\_\_\_, making routing more efficient. A label header is inserted in front of the IP header.

【解答】MPLS (Multiprotocol Label Switching)。 [ch10 p52–56]

8. IPv6 extension headers are daisy chained based on the \_\_\_\_ field.

【解答】Next Header。 [ch08 p93–95]

9. A \_\_\_\_ access point, operating at very high power, impersonates a real AP and does man-in-the-middle.

【解答】Evil Twin (惡意雙生 AP ; 以高功率假冒真 AP 進行中間人攻擊)。 [ch07 p24,32–33]

### B. 簡答 (Short Answer)

1. 802.11 operates in 2.4GHz and 5 GHz bands.

(1). Which band provides more non-overlapping channels?

(2). Which band is worse in radio propagation?

【解答】(1) 5 GHz (11–24 個非重疊通道 ; 2.4 GHz 只有 3 個 : 1,6,11)。 (2) 5 GHz (頻率較高、傳播較差 ; 2.4 GHz 傳播特性較好)。 [ch06 p34,37]

2. The five wireless propagation problems ...

(1). What is Multipath Interference?

(2). Which two problems get worse as frequency increases?

(3). Which is the most serious propagation problem at WLAN frequency?

【解答】(1) 直接波與反射波同時到達接收端而互相干擾；若兩波反相會抵消成無訊號。(2) Shadow zones(死區) 與 Absorptive attenuation(吸收衰減) 隨頻率升高而變嚴重。(3) Multipath Interference(多路徑干擾)。〔ch06 p17-22〕

3. NCNU has three sites (Taichung, Tainan, Taipei) to be connected via WANs. ChungHwa Telecom offers leased lines and a PSDN with POPs in the three cities. Recommend a WAN solution with a figure.

【解答】建議用 PSDN (如 MPLS/Carrier Ethernet)：每個站點只需「一條專線」當 access line 連到當地最近的 POP，再由電信業者的 PSDN 雲把三站互連。比起三站兩兩拉專線(全網狀需 3 條、N 站需  $N(N-1)/2$  條)成本低很多，且具 QoS。圖示：台中／台南／台北三站各→專線→各自 POP→中華電信 PSDN 雲(互連)。〔ch10 p33-37,47-56〕

4. (1) SYN means synchronize—what is synchronized? (2) What is the four-way closing? (3) Why four steps?

【解答】(1) 同步雙方的「初始序號(initial sequence number)」。(2) 四次揮手關閉：FIN→ACK→FIN→ACK，兩個方向各自關閉。(3) TCP 是全雙工，兩個傳輸方向獨立；一方送完(FIN)時另一方可能還在傳，所以每個方向各用一組 FIN/ACK 分開關閉，共四步。〔ch08 p104,109〕

5. CSMA/CA+ACK is the mandatory MAC method of 802.11.

(1). What is CS? (2). What is CA? (3). What is ACK?

【解答】(1) CS(Carrier Sense)：傳送前先聆聽載波，忙碌就等待。(2) CA(Collision Avoidance)：通道空出後，各等待裝置先等「隨機」退避時間再傳，避免同時搶送碰撞。(3) ACK：接收端正確收到後立即回送確認；發送端收不到 ACK 就重送→使 802.11 成為可靠協定。〔ch06 p59-61〕

6. PSK and 802.1X are the two 802.11i modes.

(1). Which needs a central authentication server?

(2). Explain “In PSK mode, pre-shared key is shared, but the session key is unshared.”

【解答】(1) 802.1X 模式。(2) 所有站台共用同一把 PSK (用於初始認證)，但每台站台再由它各自推导出「不同的」每次連線專屬 session key 來實際加密，因此不同站台的加密(session)金鑰並不相同。〔ch07 p13-16〕

7. Explain (1) SQL Injection and (2) Cross-Site Scripting.

【解答】(1) SQL Injection：在輸入欄填入 SQL 片段使查詢語意被竄改，例如輸入 ' OR '1'='1 使 WHERE 永遠為真，繞過登入或竊取資料。(2) XSS：把腳本(如 `<script>alert(document.cookie)</script>`)注入會被網站回填到網頁的輸入；受害者瀏覽器執行該腳本(如竊取 cookie)。〔ch11 p7,9,10〕

8. Routing table (metric = speed).

| Row | Destination | Subnet mask | Metric (speed) | Interface | Next Hop     |
|-----|-------------|-------------|----------------|-----------|--------------|
| 1   | 0.0.0.0     | 0.0.0.0     | 6              | 1         | 163.22.1.254 |

|    |             |               |    |   |              |
|----|-------------|---------------|----|---|--------------|
| 2  | 140.0.0.0   | 255.0.0.0     | 10 | 6 | 140.1.1.254  |
| 3  | 140.20.0.0  | 255.255.0.0   | 12 | 5 | 140.20.1.254 |
| 4  | 140.20.0.0  | 255.255.0.0   | 18 | 8 | 140.19.1.254 |
| 5  | 140.36.0.0  | 255.255.0.0   | 12 | 9 | 140.22.2.1   |
| 6  | 140.36.24.0 | 255.255.255.0 | 25 | 2 | 140.21.3.254 |
| 7  | 140.37.0.0  | 255.255.0.0   | 12 | 3 | 140.25.1.254 |
| 8  | 140.37.24.0 | 255.255.255.0 | 6  | 7 | local        |
| 9  | 140.37.26.0 | 255.255.255.0 | 26 | 4 | 140.37.2.1   |
| 10 | 140.37.26.0 | 255.255.255.0 | 30 | 3 | 140.37.3.252 |

(1). dest = 140.37.26.9 ? (2). dest = 140.36.25.7 ? (3). dest = 142.20.1.2 ?

【解答】(1) 符合 Row 1,2,7,9,10 ; 9、10 皆 /24, 比速度 30>26 → 選 Row 10。

【解答】(2) 符合 Row 1,2,5 ; 最長前綴 Row 5 為 /16 → 選 Row 5。(Row 6 為 140.36.24.0/24, 25≠24 不符)

【解答】(3) 只符合 Row 1 (142.x 不屬 140.x ; 其餘皆不符) → 選 Row 1 預設路由。〔ch08 p34-48〕

#### 9. Compare IPsec Transport mode and Tunnel mode in terms of security and cost.

【解答】Transport : 保護「主機到主機」, 涵蓋 Internet 與站內網路 ; 每台主機都需憑證與設定 → 安全性較佳但成本較高(且防火牆讀不到加密內容)。Tunnel : 只有兩端 gateway 需憑證/設定, 只保護 Internet 段、站內不保護 → 成本較低但安全性較差。結論 : 安全 Transport 勝 ; 成本 Tunnel 省。〔ch09 p48-52〕

#### 10. Give one protocol for each:

(1) VoIP signaling (2) sending e-mails (3) internet-layer error/supervisory (4) dynamic routing (5) reliable (6) used by ping (7) router finds MAC of next hop/destination

【解答】(1) SIP (2) SMTP (3) ICMP (4) OSPF (或 RIP/EIGRP/BGP 擇一) (5) TCP (6) ICMP (7) ARP。〔ch08-ch11〕

### 試卷三：企業資訊通訊與網路 期末考古題（年份不詳）

#### A. Fill in the Blanks (30%)

1. \_\_\_\_ allows increasing trunk speed by connecting two Ethernet switches using two or more trunk lines.

【解答】Link aggregation（鏈路聚合 / bonding）。〔ch05 trunk〕

2. IEEE 802 divided the data link layer into two layers: \_\_\_\_ layer and LLC layer.

【解答】MAC（Media Access Control）。〔ch08 p5〕

3. In a hierarchical Ethernet LAN, the number of paths between any two end hosts is \_\_\_\_.

【解答】one（只有一條路徑，故交換表每個位址只有一列）。〔ch08 p24〕

4. Most PSDNs select the best path before transmission; the actual flow follows this path, called the \_\_\_\_.

【解答】virtual circuit（虛擬電路 VC）。〔ch10 PSDN〕

5. By inverse square law, at 10 m strength is 80 mW; at 40 m it becomes \_\_\_\_ mW.

【解答】5 mW。  $80 \times (10/40)^2 = 80 \times 1/16 = 5$ 。〔ch06 p14–16〕

6. In cellular telephony, \_\_\_\_ occur when a subscriber moves between cells.

【解答】handoffs（換手 / roaming）。〔ch10 p26 ; ch06 p56〕

7. By Shannon equation, transmission speed can be doubled by doubling the \_\_\_\_.

【解答】bandwidth（頻寬；頻寬乘 N 速度可達 N 倍）。〔ch06 p31–32〕

8. In 802.11g, the three non-overlapping channels are \_\_\_\_.

【解答】1, 6, 11。〔ch06 p34〕

9. In the \_\_\_\_ mode of WPA and 802.11i, all stations and an AP share the same initial key.

【解答】PSK（Pre-Shared Key）。〔ch07 p13〕

10. If a router knows a target's IP address, it may use \_\_\_\_ to find the target's MAC address.

【解答】ARP（Address Resolution Protocol）。〔ch08 p60–64〕

11. If a browser gets a host name "www.abc.com", it may use \_\_\_\_ to find the IP address.

【解答】DNS（Domain Name System）。〔ch09 p27–32〕

12. A router decrements the \_\_\_\_ field of the IPv4 header by 1; if it reaches 0 the packet is dropped.

【解答】TTL（Time to Live）。〔ch08 p70〕

13. IPsec's \_\_\_\_ mode provides no security in the site network and needs no digital certificate in each host.

【解答】Tunnel（隧道模式）。〔ch09 p50–52〕

14. IETF defines \_\_\_\_ as the signaling protocol of VoIP.

【解答】 SIP (Session Initiation Protocol) 。 [ch11 p41–42]

15. If the host part is 6 bits, the maximum number of hosts in the subnet is \_\_\_\_.

【解答】 62 ( $2^6-2$ ) 。 [ch09 p6]

## B. Short Answer (70%)

1. Give an example of a switching table in an Ethernet switch.

【解答】 一張「EUI-48(MAC)位址 → 連接埠(port)」對照表；例如 A1-B2-C3-D4-E5-F6 → Port 3。  
交換器查目的 MAC 找到唯一輸出埠。 [ch08 p24,29]

2. (a) What is Virtual LAN (VLAN)? (b) Two benefits of VLANs?

【解答】 (a) VLAN：在交換器內以邏輯方式劃分的廣播網域，與實體位置無關，依群組(而非實體埠)歸類主機。(b) 好處：①安全/隔離—縮小廣播網域、不同 VLAN 間流量隔離；②彈性—可不受實體位置限制邏輯分群、調動方便，並減少廣播流量。 [ch05 VLAN]

3. Sort by speed ascending: 10Mbps/2Mbps ADSL, 802.11g, T1, T3.

【解答】 T1 (1.544 Mbps) < ADSL (下行 10 Mbps) < T3 (44.736 Mbps) < 802.11g (54 Mbps 額定)。  
[ch10 p41 ; ch06]

4. The golden zone for commercial mobile services is 500MHz–10GHz. Why is it "golden"?

【解答】 太低的頻率可用頻寬太少(速度受限)；太高的頻率傳播問題嚴重(吸收衰減、死區、多路徑都隨頻率變差，且需視線傳播)。500 MHz–10 GHz 兼顧「足夠頻寬(高速)」與「可接受的傳播特性」，最適合商用行動/資料服務，故稱黃金頻段。 [ch06 p11,17–22,31–32]

5. Explain (a) Multipath Interference (b) Shadow Zones (c) Absorptive Attenuation.

【解答】 (a) 多路徑干擾：直接波與反射波互相干擾，反相時抵消；WLAN 頻段最嚴重。(b) 死區/陰影區：障礙物擋住訊號無法到達的區域，頻率越高越嚴重。(c) 吸收衰減：訊號能量被空氣/水/障礙物吸收而變弱，頻率越高越嚴重。 [ch06 p17–22]

6. (a) Describe CSMA/CA+ACK (b) Describe RTS/CTS.

【解答】 (a) CSMA/CA+ACK：傳送前聆聽載波(CS)，忙就等待；空出後各等隨機退避再送(CA)避免碰撞；接收端正確收到回 ACK，沒收到 ACK 就重送(可靠)。802.11 強制使用。(b) RTS/CTS：選用機制；傳送前先送 RTS、AP 回 CTS，所有聽到 CTS 的站台都先暫停，藉此解決「隱藏節點(彼此聽不到)」問題。 [ch06 p59–61,69]

7. Compare 802.11, Bluetooth, and UWB in speeds and distance limits.

【解答】 802.11(Wi-Fi)：速度高(數十 Mbps~Gbps)、距離較遠(WLAN，約 ~100 m)。Bluetooth：速度低(經典 ~3 Mbps、高速版 ~24 Mbps)、距離短(~10 m，高速版 ~30 m)，屬 PAN。UWB：以極寬頻寬達很高速度(數百 Mbps)，但距離很短(約 ~10 m)。 [ch07 p50–52]

8. (a) Three-way opening of a TCP connection (b) Four-way closing.

【解答】 (a) 三次握手開啟：SYN → SYN+ACK → ACK (同步雙方初始序號後建立連線)。(b) 四次揮手關閉：FIN → ACK → FIN → ACK (全雙工，兩方向各自關閉)。 [ch08 p104,109]

9. Routing table (metric = cost).

| row | Destination    | Subnet mask     | metric | Interface | Next Hop      |
|-----|----------------|-----------------|--------|-----------|---------------|
| 1   | 0.0.0.0        | 0.0.0.0         | 10     | 2         | 163.22.1.254  |
| 2   | 140.128.0.0    | 255.255.0.0     | 8      | 3         | 140.128.1.254 |
| 3   | 140.128.12.0   | 255.255.255.0   | 7      | 7         | 140.128.2.1   |
| 4   | 140.128.12.128 | 255.255.255.192 | 12     | 5         | local         |
| 5   | 140.128.15.0   | 255.255.255.0   | 8      | 4         | 140.128.3.253 |
| 6   | 140.128.15.0   | 255.255.255.0   | 6      | 6         | 140.128.3.254 |
| 7   | 140.128.15.16  | 255.255.255.255 | 11     | 1         | local         |
| 8   | 140.128.18.0   | 255.255.255.0   | 9      | 3         | 140.128.1.253 |
| 9   | 140.128.18.64  | 255.255.255.240 | 12     | 1         | 140.128.4.254 |

(a). 140.128.12.80 (b). 140.128.12.140 (c). 140.128.15.200

【解答】(a) 符合 Row 1,2,3 ; 最長前綴 Row 3 為 /24 → 選 Row 3。(.80 & /26 = .64 ≠ .128, Row 4 不符)

【解答】(b) 符合 Row 1,2,3,4 ; Row 4 為 /26 最長(.140 & /26 = .128 符合) → 選 Row 4。

【解答】(c) 符合 Row 1,2,5,6 ; 5、6 皆 /24, metric=cost 取小 6<8 → 選 Row 6。(Row 7 為 .16/32 不符) [ch08 p34-48]

10. Which are private addresses: 10.0.0.3, 19.16.0.1, 72.168.1.1, 100.172.0.1, 192.168.10.201?

【解答】私有位址：10.0.0.3 (10/8) 與 192.168.10.201 (192.168/16)。其餘為公有(19.x、72.x、100.x 皆不在三段私有範圍 10/8、172.16-31、192.168/16 內)。[ch09 p14,25]

11. Give an example of an HTTP request message.

【解答】GET /panko/home.htm HTTP/1.1[CRLF] / Host: voyager.shidler.hawaii.edu / Connection: Keep-Alive。[ch11 p51]

12. Describe NAT operation in terms of the translation table.

【解答】外送封包：NAT 防火牆把內部來源 IP+port 換成對外(假)IP+port，並把對應關係(內部 IP:port ↔ 對外 IP:port)記入轉換表。回送封包：依轉換表反查還原成原本的內部 IP:port。對內外主機都透明，可擴充位址數並增加安全性。[ch09 p16-21]

13. Briefly explain: (a) MTSO (b) SMTP (c) SaaS (d) 1000Base-T.

【解答】(a) MTSO：行動電話交換局，連接各 cellsite 並接到 PSTN，負責換手與通道分配。(b) SMTP：簡單郵件傳輸協定，在郵件伺服器間遞送(寄)電子郵件。(c) SaaS：軟體即服務，應用軟體以雲端服務方式提供，使用者經網路(瀏覽器)使用。(d) 1000Base-T：Gigabit Ethernet(1 Gbps)，跑在 4 對 UTP 銅纜上、最長 100 m。[ch10 p28 ; ch11 p13]